

1. **What is the main goal when optimizing batch size in DataLoader?** 1 point
 - ☐ Always use the largest possible batch size to maximize efficiency
 - ☒ Find a balance that maximizes GPU efficiency while fitting within memory constraints
 - ☐ Use the smallest batch size to reduce memory usage
 - ☐ Set batch size equal to the number of workers
2. **What is pinned memory in PyTorch DataLoader?** 1 point
 - ☐ Memory that is compressed to save space
 - ☐ Memory that is shared between CPU and GPU
 - ☒ A special region of RAM that stays fixed in place, allowing faster GPU data transfer
 - ☐ Memory that is automatically freed after each batch
3. **What does the prefetch_factor parameter control in PyTorch DataLoader?** 1 point
 - ☐ The number of workers to use for data loading
 - ☒ How many batches each worker preloads into memory ahead of time
 - ☐ The size of each batch
 - ☐ Whether to use pinned memory or not
4. **What is the default value of prefetch_factor in PyTorch DataLoader?** 1 point
 - ☐ 1
 - ☒ 2
 - ☐ 4
 - ☐ Equal to the number of workers
5. **What is the primary goal when optimizing training loops?** 1 point
 - ☐ Increase model accuracy at any cost
 - ☒ Make models train faster and more efficiently without sacrificing accuracy
 - ☐ Use as much GPU memory as possible
 - ☐ Reduce the number of training epochs
6. **Which of the following questions can the PyTorch Profiler help answer?** 1 point
 - ☐ What is the optimal learning rate for my model?
 - ☒ Which operations are consuming the most time during training?
 - ☐ How many epochs should I train for?
 - ☐ What batch size will give the best accuracy?
7. **In the Lightning profiler configuration, what does the schedule parameter with wait=2, warmup=2, active=10 mean?** 1 point
 - ☒ Wait 2 epochs with the profiler inactive, warm up for 2 epochs with the profiler active, then profile for 10 epochs
 - ☐ Wait 2 batches with the profiler inactive, warm up for 2 batches with the profiler active, then record performance data for 10 batches
 - ☐ Use 2 workers, 2 warmup steps, and profile 10 models
 - ☐ Profile every 2nd batch for 10 total measurements
8. **What does mixed precision training primarily aim to achieve?** 1 point
 - ☐ Increase model accuracy by using higher precision
 - ☒ Lower memory usage and faster computation while maintaining stability
 - ☐ Reduce the number of parameters in the model
 - ☐ Eliminate the need for gradient computation
9. **How does gradient accumulation work?** 1 point
 - ☒ It stores gradients from previous epochs to speed up training
 - ☐ It automatically adjusts the learning rate based on gradient magnitude
 - ☐ It reduces the number of gradients computed per batch
 - ☐ It accumulates gradients over multiple smaller batches before performing an optimizer step
10. **What is the main advantage of Lightning's architectural approach compared to standard PyTorch?** 1 point
 - ☐ Lightning runs faster than PyTorch
 - ☐ Lightning uses less memory than PyTorch
 - ☒ Lightning forces a clean separation between model logic and training infrastructure
 - ☐ Lightning only works with specific model types